

Kirlous Helal Zaki

Flutter Developer

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CAREER OBJECTIVE

Motivated Computer Science graduate and aspiring Flutter Developer, with hands-on experience building cross-platform mobile applications for Android and iOS using Flutter, Bloc/Cubit, and clean architecture design patterns. Skilled in Agile/Scrum environments, version control (Git), and problem-solving. Reliable and detail-oriented, eager to contribute to a collaborative engineering team building scalable, well-tested mobile solutions.

EDUCATION

Bachelor's Degree in Computer and Information Science, Jul 2024
Ain Shams University
Scientific Computing Department, Cumulative GPA 2.96/4.0 (Very Good)

PROFESSIONAL EXPERIENCE

AI Image Classification & Enhancement System (Python), Jul 2024
Freelance collaboration with a PhD researcher

- Trained and integrated multiple CNN/ML models into a single pipeline to predict digital-print quality factors (color type, contrast, resolution) from an image
- Built a Tkinter GUI with image enhancement features (CMYK conversion, contrast, EDSR x3 upscaling) and packaged it as a standalone .exe

Internship Automation and Manual API Testing, (Certificate) Jul 2024

- Gained experience in validating API functionality manually.
- Learned to automate test scripts for faster defect detection.

MEAN Stack ITI Summer Code Camp, (Certificate) Aug 2022

- Developed front-end and back-end web applications using the MEAN stack.
- Integrated MEAN stack components and gained experience in building scalable web applications.

Git/Github Summer Workshop Sep 2022

- Applied Git and GitHub for version control, change tracking, and collaborative software development.

COURSES

Flutter Course Udemy Nov 2024

Implementing and Developing SQL server objects Mar 2025

Asp.Net Core Web Api Sep 2026

CERTIFICATES

Competitions

- Benha University Competition Certificate — Cardiac Function Assessment Graduation Project
- Computer Vision Competition Certificate — Multi-Cancer Classification Project
- Machine Learning Competition Certificate — Hotel Rating Prediction Project

Courses & Training

- Graduation from College of Computer Science, Ain Shams
- Flutter Course Certificate
- Internship Testing Certificate
- MEAN Stack Development Course (ITI)
- Military Service Certificate

PROJECTS

Cardiac Function Assessment(GP – Flutter & Deep Learning)

- **Problem Definition:** The left ventricle is crucial for pumping oxygenated blood, and its failure can lead to heart failure or cardiac arrest. Cardiomyopathy impacts its function, requiring early diagnosis, with echocardiograms assessing Left Ventricle Ejection Fraction (LVEF) as a key indicator of heart health.
- **Objective:** Developed a deep-learning system for echocardiogram analysis, achieving **91.8% DSC** in left-ventricle segmentation and **1.69 MSE**.
- **Competition:** Achieved **3rd Place** in Benha University Innovations Marathon. (Competition Certificate) 

Social App (Mobile App - Flutter)

- **Feature-Rich Social Platform:** Offers functionalities like user authentication, profile management, post sharing, and real-time interactions, creating an engaging social networking experience.
- **Firebase Integration:** Utilizes Firebase for backend services, including real-time database, authentication, and cloud storage, ensuring seamless and scalable performance.

Shop App (Mobile App - Flutter)

- **E-Commerce Features:** Implemented **4+ core features**, including authentication, product browsing, cart management, and checkout.
- **API Integration:** Integrated RESTful APIs for dynamic product data and responsive application updates.

Hotel Rating Prediction (Machine Learning)

- The project applies various machine learning algorithms for tasks like classification, regression, or clustering, focusing on building predictive models. It demonstrates hands-on problem-solving with real-world data, showcasing skills in model development and evaluation. (Competition Certificate) 

Multi-Cancer Classification (Computer Vision)

- **Medical Image Classification:** Developed a computer vision system using machine learning and deep learning to classify medical images for multiple cancer types, applying image preprocessing and feature extraction techniques.
- **Model Performance:** Evaluated **5 models**, achieving **97.7% test accuracy** with Gradient Boosting, **96.5%** with AdaBoost, and **91.0%** with a custom CNN. (Competition Certificate) 

Classical Encryption Techniques Package

- The project implements classical encryption techniques, including substitution ciphers (Caesar, Playfair, Polyalphabetic, Hill, Monoalphabetic) and transposition ciphers (Rail Fence, Columnar).

Cloud-Computing-Project (Docker & Kubernetes)

- Dockerized an existing e-commerce web application using **5 Docker images**, orchestrated via docker-compose and Kubernetes with **5 pods**, replica sets, and YAML deployment manifests.

SKILLS

Technical Skills:

- Flutter, Dart
- Bloc, Cubit
- Firebase (Auth, Firestore, Storage, etc)
- RESTful APIs
- SQLite, Hive
- SQL, NoSQL
- flutter_test, mockito, widget testing
- C++, C#, Java, Python, JavaScript, Html, Css
- Rest Assured

Soft Skills:

- Team Collaboration
- Analytical Skills
- Communication
- Leadership
- Problem Solving
- Work Independently
- Innovation
- Adaptability
- Project Management
- Research
- Agile/Scrum

Languages:

- Arabic – Native Proficiency
- English – Professional Working

Tools & Platforms:

- Git & GitHub (Version Control)
- Visual Studio, Android Studio
- VS Code, Cursor, Windsurf
- Postman
- Docker
- Linux
- Katalon Studio
- Selenium
- Jupyter Notebook

Core Concepts:

- MVVM Architecture
- Clean Architecture
- Repository Pattern
- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- Design Patterns & Architecture
- Artificial Intelligence, Machine Learning, Deep Learning
- Database Management
- API Design
- Testing Principles
- Network & Cybersecurity Fundamentals
- .NET Core Backend
- Angular Frontend